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🧠 Multi-Agent Systems Quiz

📌 Section A: MCQs (Conceptual Understanding)

1. What is the primary difference between a single-agent system and a multi-agent system?
A. Single-agent systems use AI, multi-agent systems do not
B. Multi-agent systems involve multiple interacting agents
C. Single-agent systems are always faster
D. Multi-agent systems cannot make decisions independently

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2. In a multi-agent system, agents are typically:
A. Completely dependent on each other
B. Independent but capable of interaction
C. Controlled by a central authority only
D. Static and non-adaptive

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3. Which of the following is an example of a multi-agent system?
A. Calculator app
B. Self-driving car network communicating with other cars
C. Text editor
D. Single chatbot

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4. What is a key advantage of multi-agent systems?
A. Simpler design
B. No communication required
C. Scalability and parallelism
D. Limited decision-making

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5. Which architecture uses only one decision-making entity?
A. Distributed system
B. Multi-agent system

C. Single-agent system
D. Hybrid system

📌 Section B: Short Answer Questions

6. Define a multi-agent system in your own words.

A multi-agent system is a system where multiple independent agents interact and collaborate to solve a problem or achieve a goal.

7. List two advantages of multi-agent systems over single-agent systems.

Better scalability

Parallel task execution (faster performance)

8. What is agent communication, and why is it important?

Agent communication is the exchange of information between agents.
It is important because it enables coordination, decision-making, and collaboration.

9. Name two types of agent communication patterns.

Broadcasting

Peer-to-peer communication

10. What is task delegation in multi-agent systems?

Task delegation is when one agent assigns tasks to other agents based on their abilities or roles.

📌 Section C: Analytical / Application-Based Questions

11. Compare single-agent and multi-agent systems based on:

- Decision-making
- Scalability
- Complexity

Feature	Single-Agent System	Multi-Agent System
Decision-making	Centralized	Distributed
Scalability	Limited	High
Complexity	Simple	More complex

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12. Explain the difference between **centralized** and **decentralized** multi-agent architectures.

Centralized: One main controller makes decisions

Decentralized: Multiple agents make independent decisions and coordinate

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13. A group of delivery robots must coordinate to deliver packages efficiently.

- What type of system is this?

Multi-agent system

- Why is communication important here?

Communication is important to:

- Avoid collisions
- Optimize delivery routes
- Share task status

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14. If one agent assigns tasks to others based on their capabilities, what strategy is being used?
A. Broadcasting
B. Task delegation
C. Isolation
D. Random assignment

15. Explain how **peer-to-peer communication** works in multi-agent systems.

Peer-to-peer communication means agents directly communicate with each other without a central controller.

📌 Section D: Scenario-Based Questions

16. In a smart traffic system, traffic lights (agents) communicate with each other to reduce congestion.

- Identify the communication pattern used

  Pattern: **Peer-to-peer communication**
- What could happen if communication fails?

  If communication fails:

- Traffic congestion increases
- Signals become inefficient
- Possible accidents

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17. Consider a team of AI agents solving a complex problem:

- One agent plans
- Others execute tasks

What architecture or strategy is this? Explain.

This scenario represents a **hierarchical multi-agent architecture**, where one agent functions as a central planner and the others act as executors. The planner agent is responsible for analyzing the overall problem, breaking it down into smaller tasks, and assigning these tasks to different agents based on their capabilities. The executor agents then carry out the assigned tasks independently and may report their progress or results back to the planner. This approach combines centralized decision-making with distributed execution, making it effective for handling complex problems in an organized and efficient manner. However, it can also create a dependency on the planner agent, meaning that if the planner fails, the entire system's performance may be affected.

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18. A multi-agent system uses broadcasting to share information.

- What are the advantages and disadvantages of this approach?

Advantages:

- Fast information sharing
- Simple implementation

Disadvantages:

- Network overload
- Redundant communication

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19. In a distributed AI system, each agent has partial knowledge of the environment.

- How can agents collaborate effectively?

Agents can collaborate by:

- Sharing local knowledge
- Communicating updates
- Using coordination protocols

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20. Suppose a system fails when one central controller crashes.

- What type of architecture is this?

Type: **Centralized architecture**

- Suggest a better alternative.

Better alternative: **Decentralized system** (no single point of failure)

📌 Section E: Higher-Order Thinking

21. Design a simple multi-agent system for **online food delivery**.

- Define roles of at least 3 agents
- **Order Agent** → handles customer orders
- **Restaurant Agent** → prepares food
- **Delivery Agent** → delivers food

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22. Why is task delegation important in large-scale systems like cloud computing?

Distributes workload

Improves efficiency

Uses resources optimally

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23. Compare **cooperative vs competitive agents** with examples.

Type	Description	Example
Cooperative	Agents work together	Delivery robots coordinating
Competitive	Agents compete	Stock trading bots

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24. How does communication overhead impact system performance?

Slows system performance

Uses more bandwidth

Causes delays

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25. Suggest improvements for a poorly coordinated multi-agent system.

Better communication protocols

Clear role assignment

Use shared memory/system

Add coordination strategies

Reduce redundant communication